Digital Image Processing Lab (NCSH 302) **Session: Monsoon 2026-27**
B. Tech. 5th Sem (CSE-Hons) **Date: 06.10.2026**

Assignment-8 (Image Segmentation)

1. **Part A.** Read "*noisy_horse.png*" as the input image f. Smooth the image using a suitable 5×5 averaging mask to obtain f_s. Replicate the border pixels while applying the mask and round the resultant values. Apply basic global thresholding to both f and f_s. Start with the average gray level as the initial threshold T. Repeat the process until the change in T is less than 0.5. Apply semi-thresholding on f_s using the following rule for thresholding with same threshold T:

$$g(f) = \begin{cases} f, & if\ f \geq T, \\ 0, & if\ f < T. \end{cases}$$

Display input image f, smoothed image f_s, the histograms of f_s with their thresholds marked and the three resultant thresholded images. *Do not use library functions for finding threshold or applying thresholding.* **[10 Marks]**

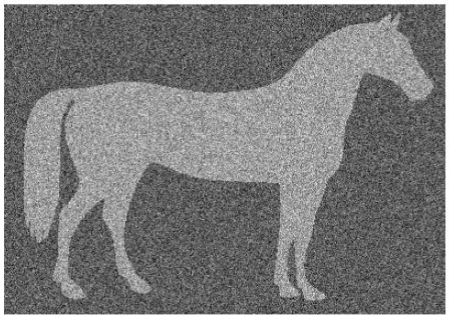
Part B. Read "*cameraman.png*" and smooth it using same averaging mask to get smoothed image f_s. Find the histogram n_z of the smoothed image, f_s. A grey level z is considered a peak if n_z is the largest value in the range $[z-20,\ z+20]$, smallest z in case of a tie. Valley m between peaks $k < l$ is the grey value with the smallest n_z in $[k, l], n_z$ is normalized by the number of pixels and use $t_{sl} = 5 \times 10^{-5}$. For every pair of neighbouring peaks - find the valley m between the two peaks, the slopes s_1 and s_2 and class of the pair (bimodal, peak and shoulder, or flat). Using the two valleys $T1 < T2$, apply

a) Multithresholding with $D_1 = [0,\ T_1 - 1], D_2 = [T_1,\ T_2 - 1], D_3 = [T_2,\ 255]$
b) Band-thresholding using D_2.

Display the histogram with all detected peaks and valleys marked, and both the thresholded results. Report the peaks, the valleys (with s_1, s_2 and class) and the number of pixels in D_1, D_2 and D_3. **[10 Marks]**

Sample Input:

noisy_horse.png

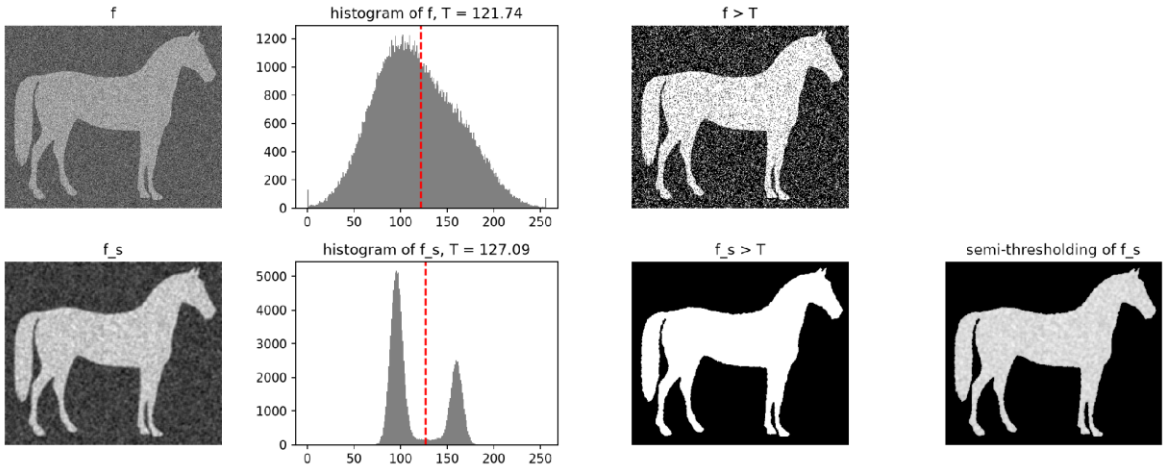


cameraman.png



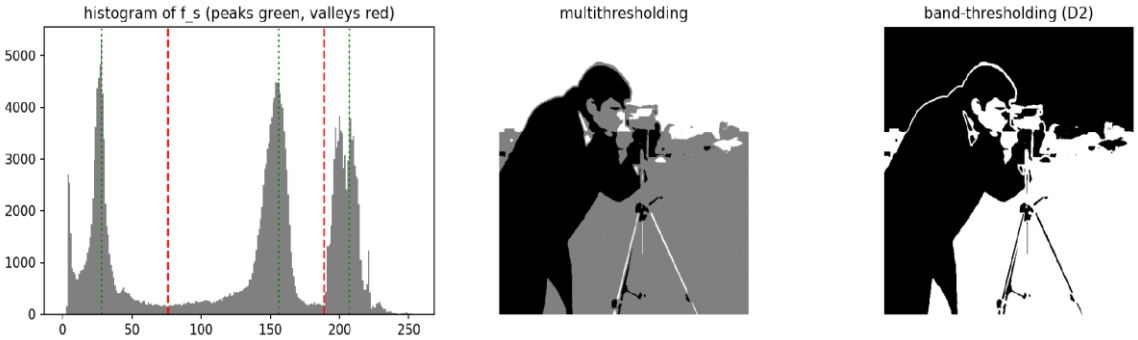
Sample Output:

Part A:



f: T = 121.74
f_s: T = 127.09

Part B:

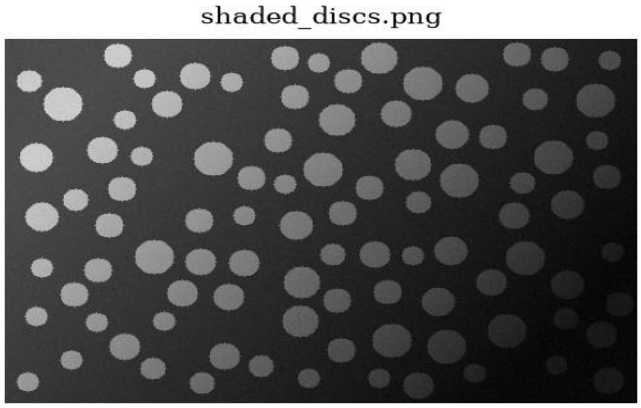


Peaks: 28, 156, 207.
Valleys: $T_1 = 76$ ($s_1 = 4.08 \times 10^{-4}$, $s_2 = 2.07 \times 10^{-4}$, bimodal),
$T_2 = 189$ ($s_1 = 4.99 \times 10^{-4}$, $s_2 = 10.74 \times 10^{-4}$, bimodal)
Pixels: $D_1 = 77{,}111$, $D_2 = 106{,}886$, $D_3 = 78{,}147$

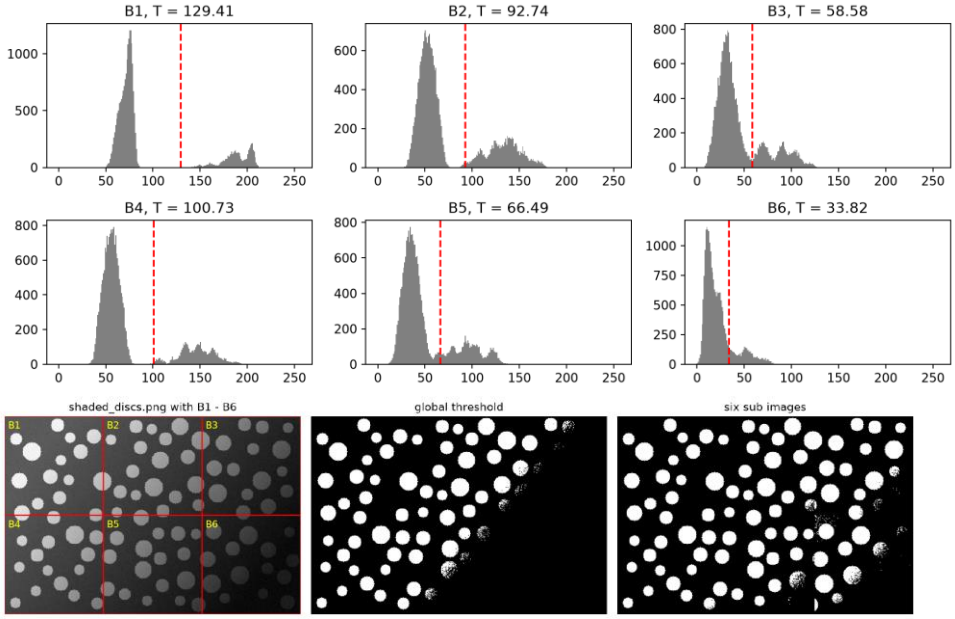
2. The image "*shaded_disks.png*" is unevenly illuminated. Read the image and apply basic global thresholding to the complete image. Divide the image into six 150×150 sub-images arranged as 2×3 grid, named B_1 to B_6 row-wise. Apply basic global thresholding separately to each sub-image. Use the average grey level as the initial threshold and repeat until the change in T is less than 0.5. For each sub-image, use the mode method to check whether its histogram is bimodal. A sub-image is considered *troublesome* if its histogram has only one peak, or the two highest peaks are not classified as bimodal by the mode method. For applying the mode method use the same previous rules used in ***Q1***. Display the histogram of each sub-image with marked thresholds, the result of global thresholding and the result after

thresholding each sub-image. Report the global T, and a tuple $(B_i, T, S_1, S_2, is_troublesome)$ for every sub-image. **[10 Marks]**

Sample Input:



Sample Output:



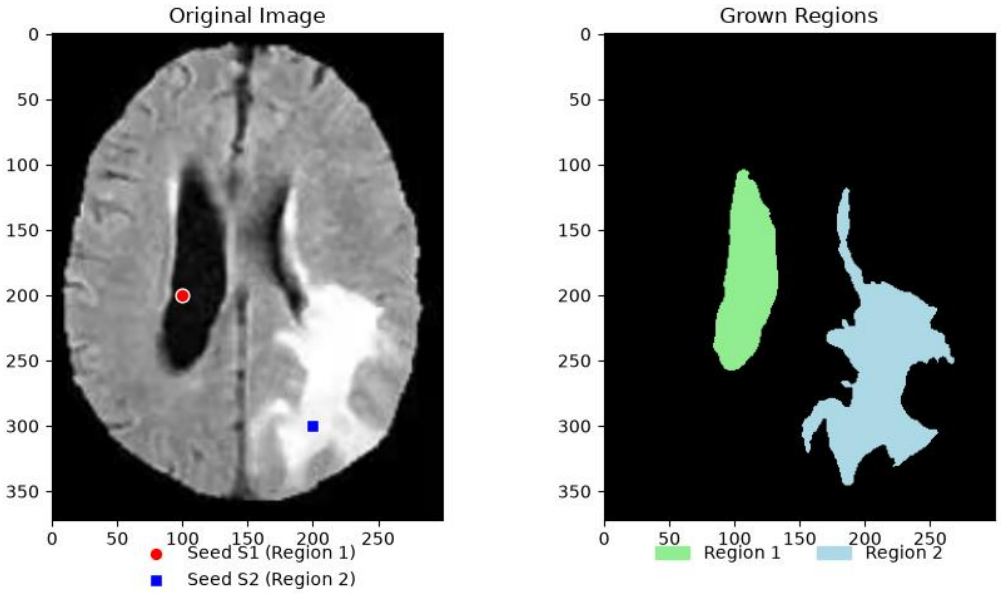
(Bi, T, S_1, S_2, is_troublesome)
Global: T = 94.26 (s1, s2 in units of 10^-4)
("B1", 129.41, 41.26, 0.82, False)
("B2", 92.74, 11.17, 1.22, False)
("B3", 58.58, 12.92, 4.19, False)
("B4", 100.73, 15.29, 1.06, False)
("B5", 66.49, 13.80, 1.73, False)
("B6", 33.82, None, None, True)
Global: T = 94.26

3. Read the image "*brain.png*" as a grayscale image and perform image segmentation using the region growing technique. Starting from the specified seed points, iteratively grow each region by including neighboring pixels that satisfy the given homogeneity criterion. Use 8-neighbour connectivity and include a neighboring pixel in the region when the following condition is satisfied: $| f(x,y) - f_8(x,y) | \leq T$, where $T = 50$, $f(x,y)$ represents the intensity of the current pixel, and $f_8(x,y)$ represents the intensity of an 8-connected neighboring pixel. Use seeds $S_1 = (200, 100)$ for Region 1 and $S_2 = (300, 200)$ for Region 2. Display the original grayscale image with seeds and the grown regions. Report the number of pixels in each grown region. **[10 Marks]**

Sample Input:



Sample Output:



Region 1: 4836 pixels
Region 2: 9269 pixels